Largen Animal Surgery and Practice

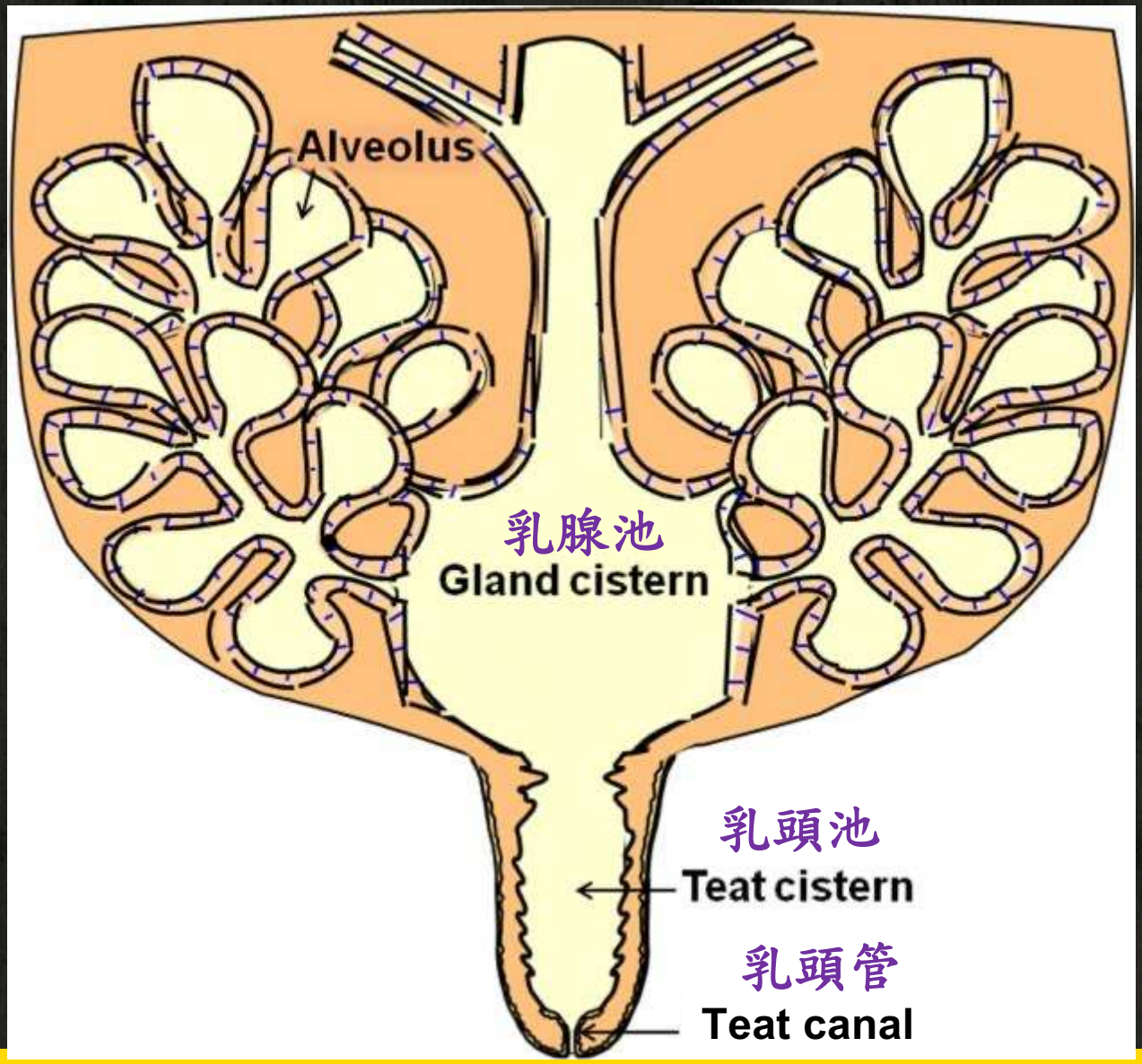
Surgery of the teat

1. Teat Lacerations



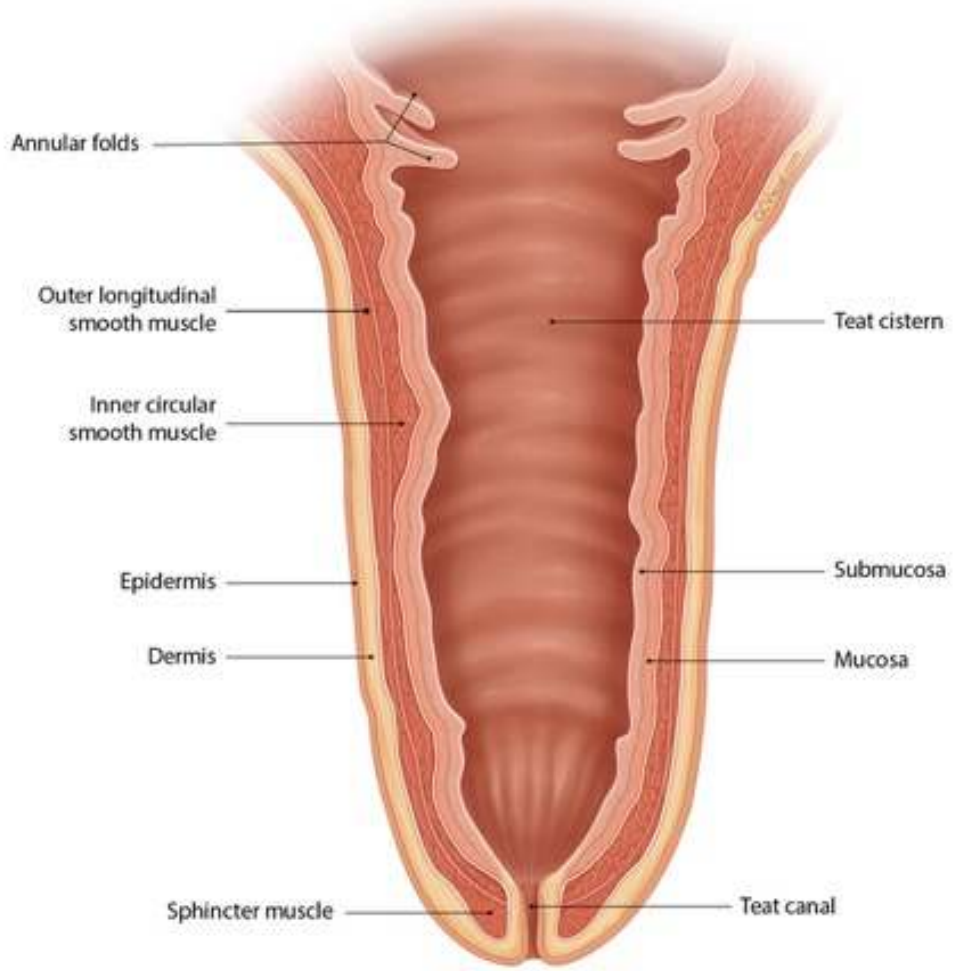


https://www.farmprogress.com/dairy-cattle/key-to-more-milk-focus-on-dairy-calf-udder-development


Alveolus
乳腺池
Gland cistern
乳頭池
Teat cistern
乳頭管
Teat canal

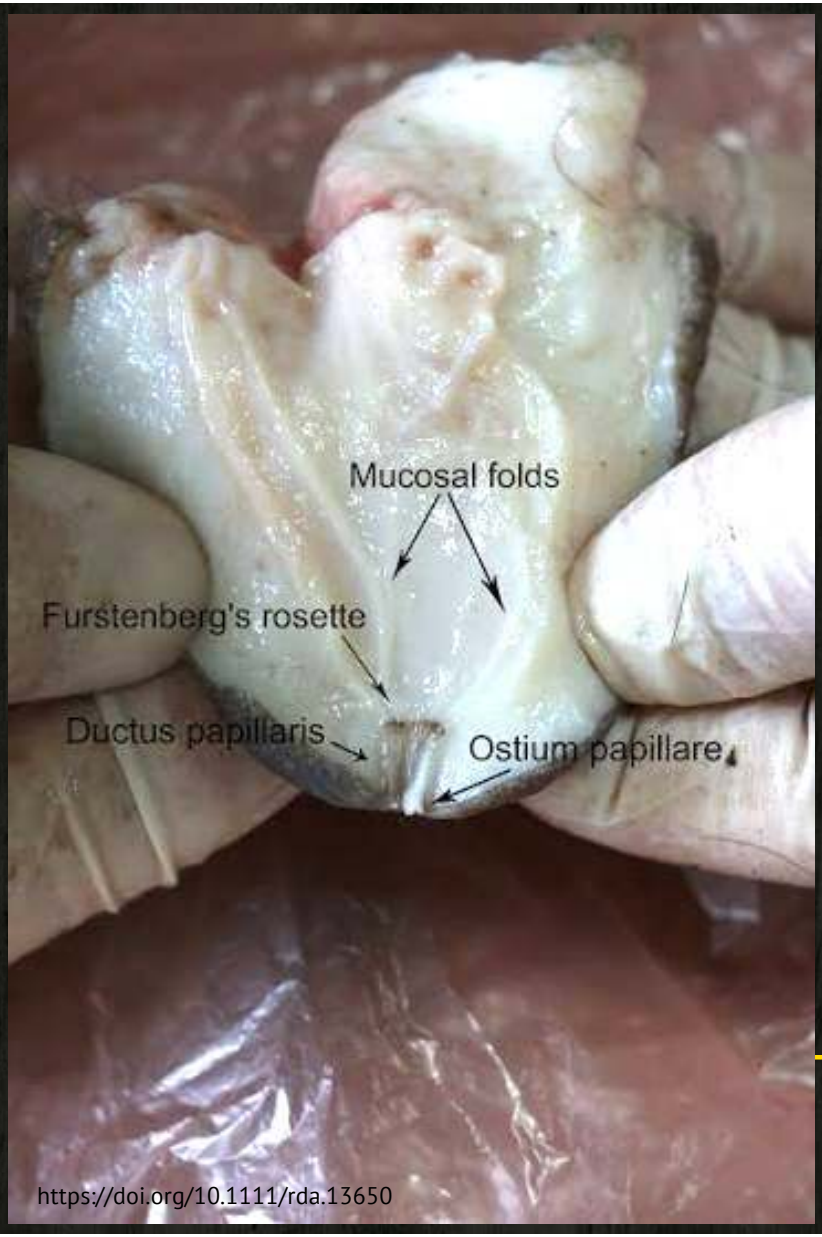


COW MAMMARY TEAT ANATOMY



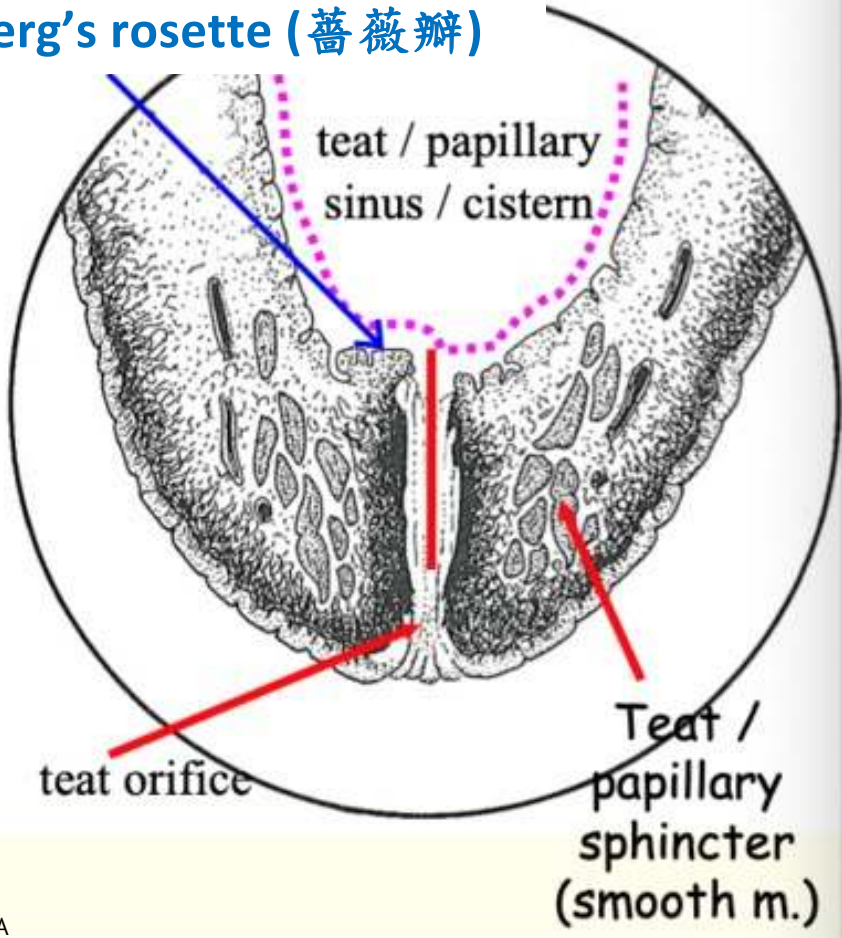
QCVISUAL Medical Illustration

https://www.qcvisual.com/portfolio.html



https://doi.org/10.1111/rda.13650

Fürstenberg's rosette (薔薇瓣)



https://reurl.cc/vGG5QA

Teat Injury

Teat injuries are common in dairy cows.

- Cut from sharp material
- Trample by the hooves (itself or others)
- Milking machine
- Lead to premature culling of dairy animals





Teat Injury

- Minor teat injuries: not involving the mucosal layer
  → Bandaging



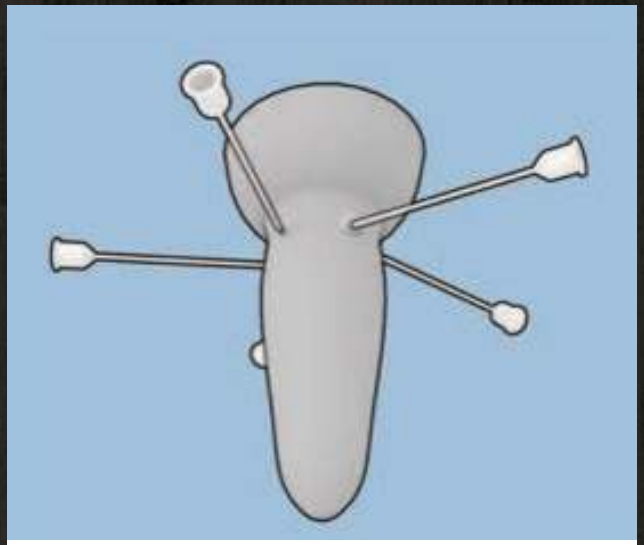
- Teat injuries extends into the mucosal layer: Suture ASAP
  - preventing bacterial infection and fistula formation

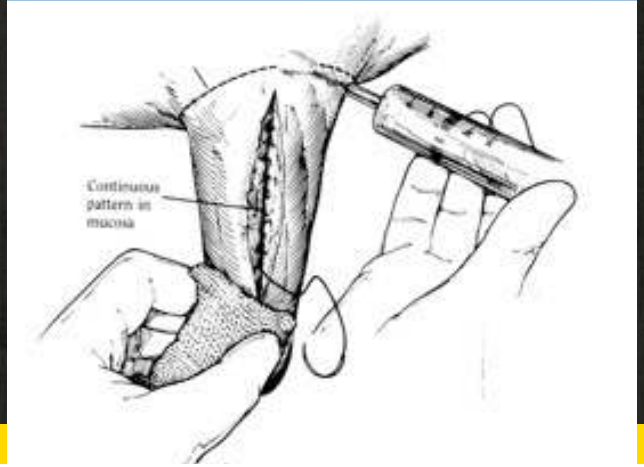
Sedation & anesthesia



- The cow should be restrained in hoof trimming chute
  - sedation if needed

Sedation & anesthesia





- The cow should be restrained in hoof trimming chute
  - sedation if needed
- Circle/Ring block: lidocaine was injected at the base of the teat
  - topical anesthesia: infused directly into the teat canal with 5–10 mL of lidocaine



Procedures

1. A rubber tourniquet is applied to the base of the teat
   - control hemorrhage and milk flow
2. The lower part of the udder and surrounding teats should be washed thoroughly
   - avoid harsh disinfectants
3. The wound edges should be freshened to remove any devitalized tissue and foreign materials
   - control hemorrhage: delay healing



Procedures

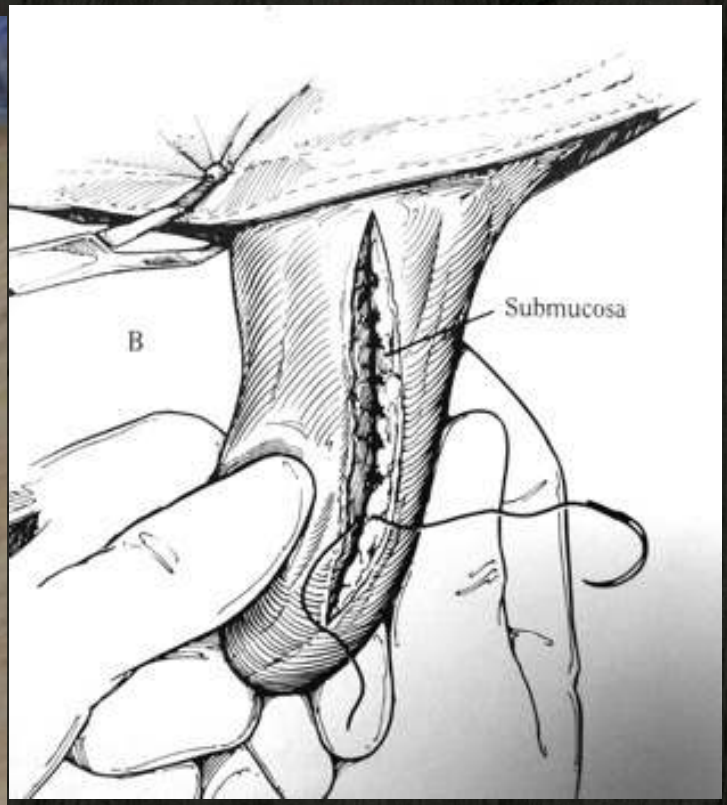
4. Closure the mucosa and muscular-submucosa separately in a simple continuous pattern, using no. 3-0 or 4-0 synthetic, monofilament, absorbable suture
   - wound edge aligned under as little tension as possible
   - Insert a teat cannula after closed the mucosa layer



https://reurl.cc/Leed9e



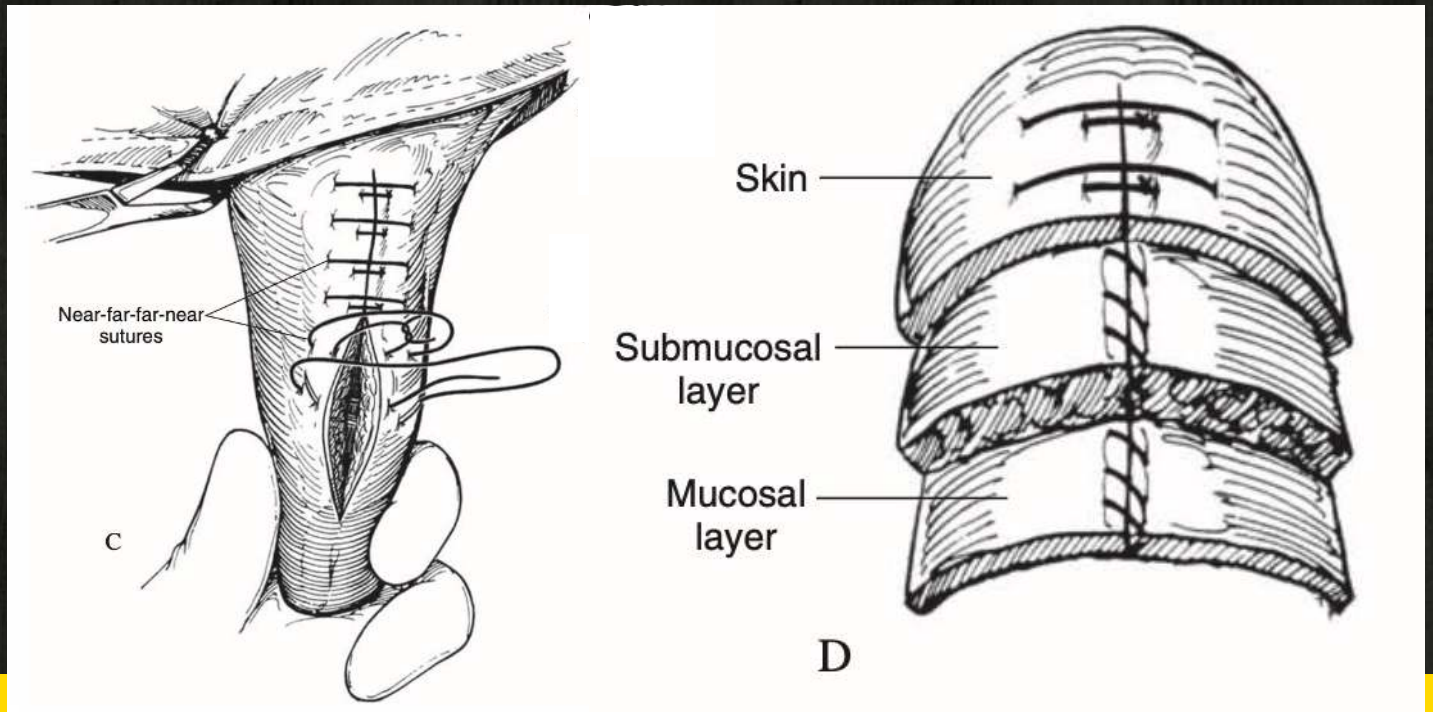




https://bvajournals.onlinelibrary.wiley.com/doi/pdf/10.1136/inp.c4574

Procedures

5. Closure the skin with a simple interrupted or near-far-far-near pattern, using no. 1-0 or 2-0 nonabsorbable suture



Procedures

6. Remove the tourniquet
7. Check for milk leakage
   - with gentle hand pressure applied to the teat



Teat fistula

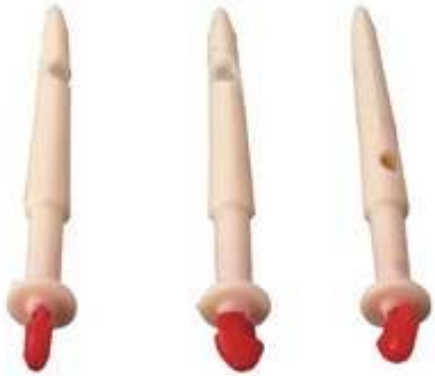
Post-operative Management

- A self-retaining teat tube (ex. Larson's teat tube) or disposable teat dilators for a week
  - keep teat canal open while tissues heal

Larson's teat tube



https://reurl.cc/3yyNol



https://reurl.cc/7YYoZy



Post-operative Management

- Use the teat cannula to allow milk drainage at the time of milking
  - do not milk by hand or machine









Post-operative Management

- Intramammary and systemic antibiotics
- Remove the suture at about 10–14 post-surgery



“

- vertical lacerations have a better prognosis
- lacerations at the distal end/ the base of the teat have poorer prognosis
- type of injury (crushing vs. linear laceration)
- degree of contamination

2.
Milk Flow Issues

No or slow milk flow!

Introduction

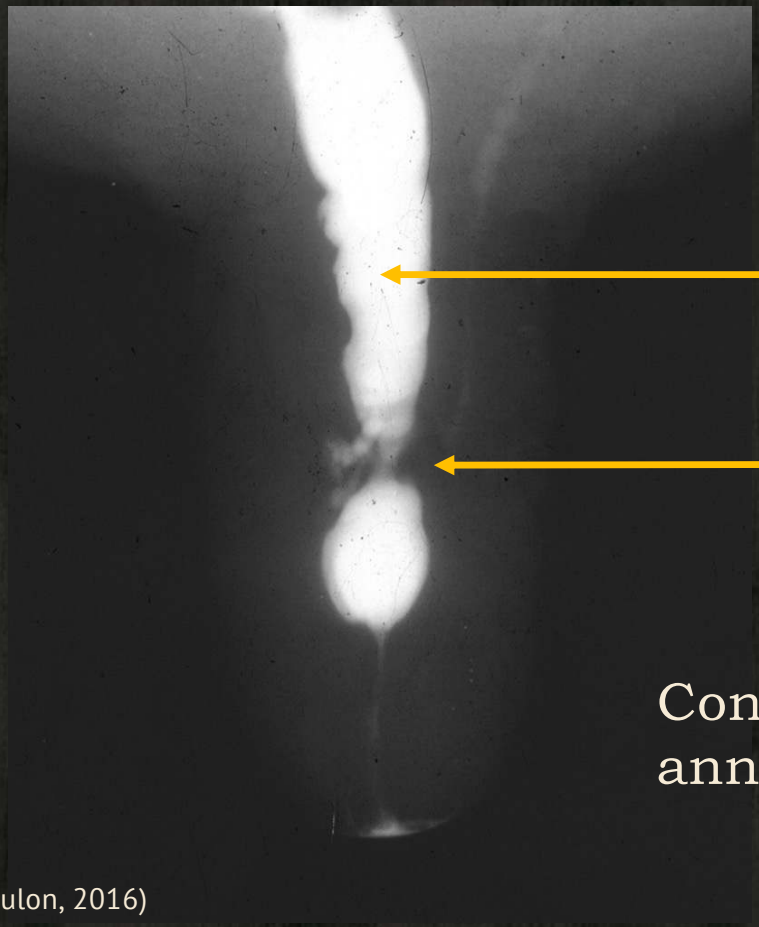
- No milk flow after parturition can occur due to trauma, obstruction from infection or milk stone or due to congenital anomalies.
  - a persistent membrane between the teat and the gland or lack of development of the teat cistern.
  - imperforate teat

Diagnosis

- Trauma, infection and milk stones are usually evident on physical examination.
- A teat cannula can be passed up the teat and used as a gentle probe to evaluate the teat cistern.
- Ultrasound, radiograph, theloscopy



Teat Stenosis



Contrast radiograph of a teat cistern with an annular fibrosis of the mucosa.
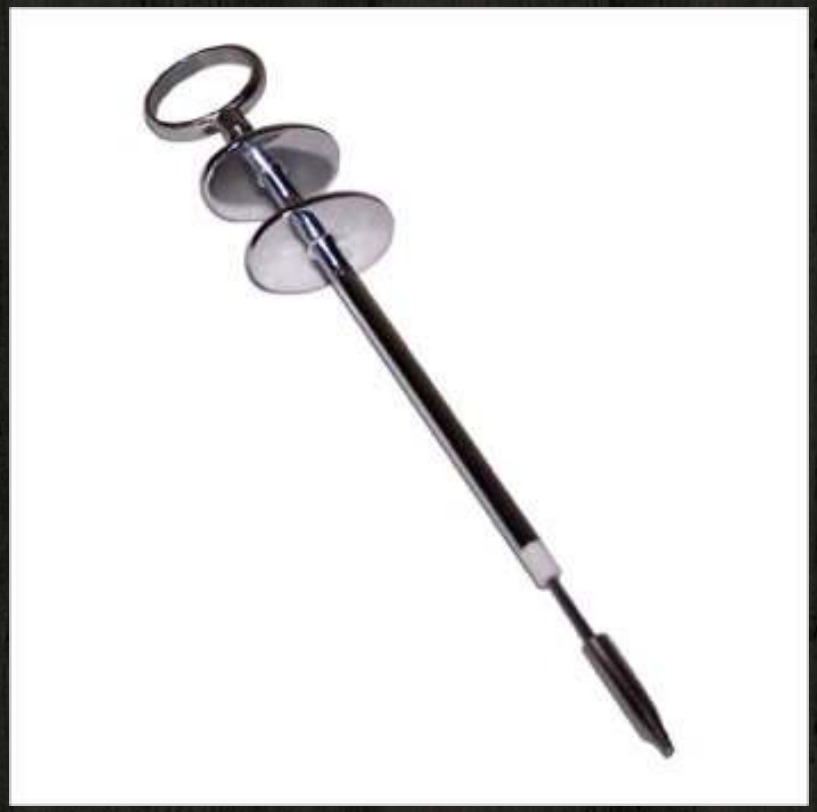
(Mulon, 2016)

Teat Stenosis

- Location: Fürstenberg rosette (47%), teat canal (35%), the middle of the teat cistern (12%) and the annular fold (6%).

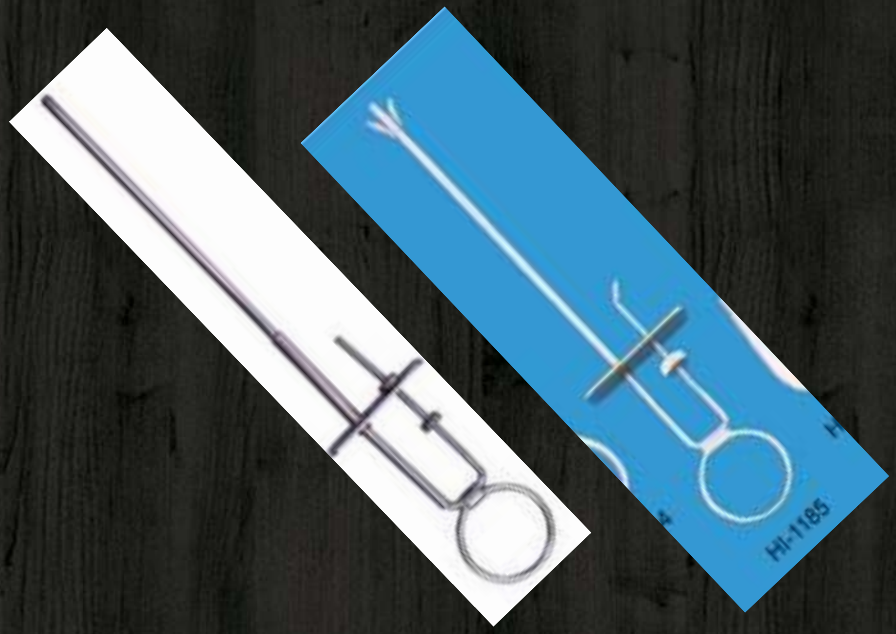
Treatment

- Milk stones are uncommon and can be removed by forceful milking or teat extractor.
- Polyps can also be removed by extractor

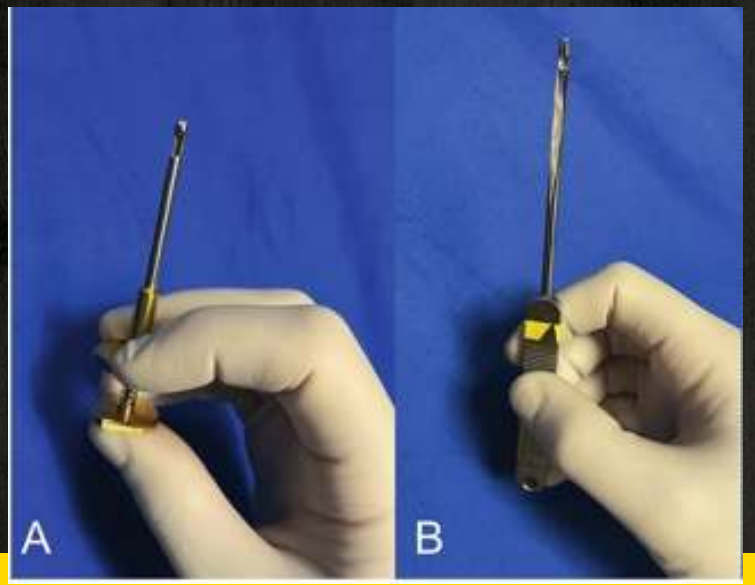


Treatment

- Membranous obstructions or the can be cut by teat slitter.



- Stenosis lesion and the polyp (growths) can be removed by Eisenhut stenosis cutter (A) or teat curette (B)



Post-operative Management

- A self-retaining teat tube (ex. Larson's teat tube) or disposable teat dilators for a week
- Use the teat cannula to allow milk drainage at the time of milking
- Intramammary and systemic antibiotics



Thank you!